

long particle accelerator (Fig. 7) is an example of a linear accelerator. Circular accelerators employ strong bending magnets to keep the particles on their circular path. Smaller the diameter of the circular path, larger and more powerful the focusing magnet needs to be.

The Large Hadron Collider (LHC) is an example of a ring accelerator (Fig. 8). It is the world's largest and most powerful particle accelerator. The LHC has a 27km ring of superconducting magnets with a number

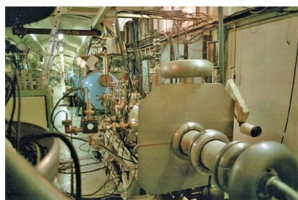


Fig. 7: Stanford linear accelerator

of accelerating structures to boost the energy of the particles along the way. Inside the accelerator, two high-energy particle beams travel in opposite directions in separate beam pipes at ultra-high vacuum, very close to the speed of light, before these are made to collide.

Two common types of linear accelerator technologies used for development of particle weapons include RF linear accelerators (RF LINACS) and induction linear accelerators (induction LINACS). Although linear accelerators are capable of accelerating particles to energy levels high enough for use as a weapon, their current-carrying ability is severely limited. As a consequence, these are not suitable for building accelerators for endoatmospheric particle beam weapons. RF LINAC can be a suitable candidate for building accelerators for exoatmospheric particle beam weapons

as space weapons don't call for very high beam power.

Both RF LINAC and induction LINAC use successively high voltages across a series of accelerating segments. Induction LINAC differs from RF LINAC in the mechanism used for generating the electric voltages within the segments of the two types of LINACS. Compared to the RF LINAC, the induction LINAC produces a more stable beam at high beam currents. The induction LINAC is therefore a more likely candidate for an endoatmospheric beam weapon.

Power source

Development of power source for an endoatmospheric particle beam weapon is a big technological challenge. The power supply is required to supply high energy over short time periods, which translates into very-high-pulsed

power levels. Advanced pulsed power technology is therefore required to develop power supply for a particle beam weapon.

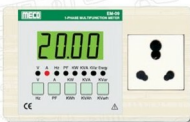
A pulsed-power device comprises three main parts: the prime power source that provides the required electrical energy over the full operating time of the weapon, energy-storage device needed for intermediate storage of the electrical energy as it is generated, and the pulse-forming network that generates power bursts or pulses of desired magnitude and time duration. Each of these three areas presents a major technological challenge.

The prime power source of a particle beam weapon is required to deliver power levels in the range of mega-watts to giga-watts, and at the same time be as lightweight and compact as possible. Also, in many applications, the prime power source needs to be mobile.

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